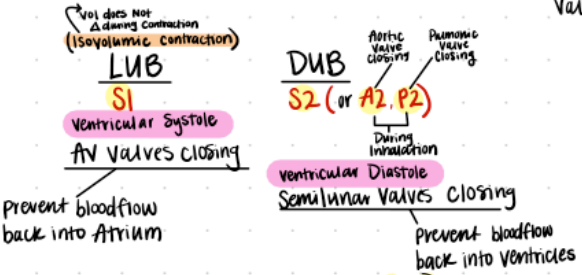
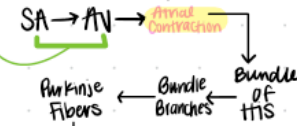
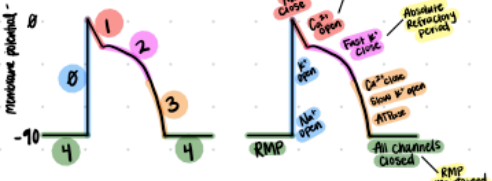


Electrical Impulse:



Cardiac RMP = -90mV



- Phase 4: RMP (more Na⁺ outside, more K⁺ inside)
- Phase 0: Depolarization
- Phase 1: Wait
- Phase 2: plateau
- Phase 3: Repolarization

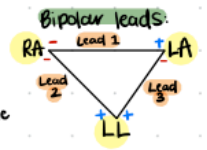
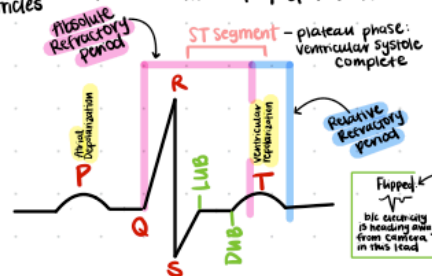
ARP (Action Repolarization Period)

RRP (Relative Refractory Period)

ECG/EKG -

recording electrical activity that then results in contraction/relaxation

Electrical impulses result from waves of the depol/repol of AP



Electrode is camera. As electricity heads toward camera, electrical potential ↑

- Unipolar leads:**
- AVR - R arm
 - AVL - L arm
 - AVF - L leg (foot)
- Chest/precordial - V₁ - V₆

Lymphocytes - Derived from stem cells in bone marrow

Primary Lymphoid Organs:

- Bone Marrow
- Thymus

Secondary Lymphoid Organs:

- Lymph nodes
- Spleen - filters blood for pathogens
- Tonsils
- Peyer's patches

Blood Reservoir: venous side. Most of the blood is here @ any given time

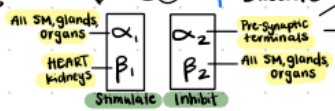
Pressure Reservoir: Arterial side. BO is the floor, Arteries provide some amt of pressure on their own

Cholinergic receptors: Bind ACh

- Muscarinic - on ♥ + vessels
- Nicotinic - on skeletal muscle

Adrenergic receptors: Bind NE

- α = Constrict
- β = Dilate



Also in blood vessels: tells them to relax

Negative Feedback! So it inhibits the post-gang neurons → prevents over exaggeration

♥ has to push ↑ higher pressure to get blood through = Raising BP

equal to vasoconstriction

Low BP: ↑ SNS activity, ↓ PSNS activity, HR ↑, TPR ↑

High BP: ↓ SNS, ↑ PSNS, HR ↓, TPR ↓

equal to vasodilation

Cerebral circulatory response is opposite SNS response elsewhere: BP ↑ → Vasoconstriction, BP ↓ → Vasodilation

Baroreceptor = Stretch receptor. Activated by ↑ BP, increases ARs to vasomotor & cardiac centers in the medulla

Chemoreceptors = respond to chemical Δ (Δ in O₂, CO₂, pH)

inverse relationship: ↑ CO₂ = ↓ pH

Cerebral circulatory response: CO₂ build-up (↑ pH) → Vasodilation, CO₂ drop (↑ pH) → Vasoconstriction

Pulse pressure = Systolic - diastolic

Stroke volume = amt of blood expelled from ♥ per systole (beat)

Normal = 30-40

Mean Arterial Pressure (MAP) = 1/3 (systolic) + 2/3 (diastolic)

to get more O₂ in



Cardiac Output = Vol pumped each min by each ventricle
 $CO = HR \times SV$

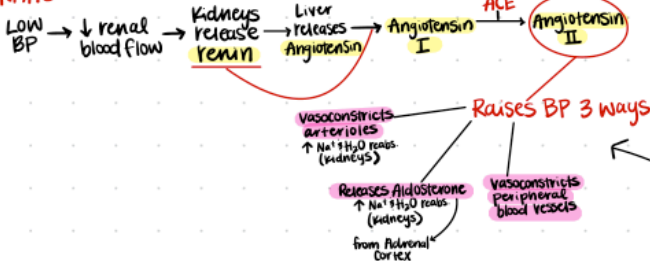
$SV = (\text{Contractility} + \text{preload}) - \text{afterload}$

Strength of contraction (inotropy) $\uparrow$
 EDV $\uparrow$
 SV $\uparrow$

Determined by venous return
 End Diastolic Vol EDV
 Vol left in ventricles @ end of diastole

TPR
 Total peripheral resistance
 TPR $\uparrow$
 SV $\downarrow$
 impedes ejection of blood from Ventricle

RAAS $\rightarrow$



BP Regulation $\rightarrow$

OSmoreceptors detect $\uparrow$ plasma osmolarity (Na^+ concn)
 $\rightarrow$ can $\uparrow$ due to $\uparrow$ Na^+ intake OR $\downarrow$ water intake
ADH (Vasopressin) released from post pituitary
 Causes $\uparrow$ **thirst**
 Stimulates H_2O reabsorption in kidneys
 $\rightarrow$ stimulates $\downarrow$ output
 Brings up blood vol $\approx$ BP

Stretch receptors in Atria of $\heartsuit$ detect $\uparrow$ blood volume

Atrial Natriuretic Peptide (**ANP**) released from atria
 Causes kidneys to excrete H_2O and Na^+ (water follows Na^+)

Stimulates $\uparrow$ output
 Brings down blood vol $\approx$ BP
 *ANP makes you pee

Aldosterone - secreted by Adrenal cortex when blood vol $\approx$ BP are reduced
 Stimulates reabsorption of Na^+ & water in the kidneys
 $\rightarrow$ stimulates $\downarrow$ urine output
 Brings up blood vol ($\uparrow$) $\approx$ BP ($\uparrow$)

Regulated by RAAS

WBC

Hematopoiesis: All blood cells start in Bone Marrow (multipotential hematopoietic stem cells)

Interleukin-3
 Controls growth & reproduction (growth inducer)

- 1) Stay in bone marrow for backup
- 2) committed stem cells (not mature yet)
- 3) Differentiate

Leukocytes:

Granulocytes - Neutrophils
 Eosinophils
 Basophils

Agranulocytes - Lymphocytes
 Monocytes $\rightarrow$ become Macrophages

Kupffer cells: Liver sinusoids macrophages
 bacteria from ingested food goes into portal blood - before entering circulation it passes through Kupffer cells to filter antigens out

Inflammation

lines of defense:

- 1st Macrophages - begin phagocytizing
- 2nd Neutrophil invasion - Byproducts of inflammation
- 3rd Macrophage invasion Again - Storage depleted
- 4th bone marrow $\uparrow$ production of granulocytes $\approx$ monocytes - committed cells stimulated

Eosinophils - parasites stimulate their production
 Also play role in inflammation
 Associated $\approx$ Allergic rxns

Basophils (in blood) - Similar to mast cells (outside the)

Both release histamine, bradykin, $\approx$ Serotonin (which makes inflammation worse)

$\rightarrow$ Mast cells more likely to release these

They both play a big role in Allergic rxns

IgE is a class of antibodies that work $\approx$ basophils & mast cells to cause allergic response

Releases a Leukotriene mixture:
 Causes anaphylaxis $\approx$ lysosomal enzymes
 $\rightarrow$ local vascular/tissue rxn (rashes)

Antigen - toxin
 Antibody - protection

Immunity:

Acquired (or adaptive) - develops $\approx$ encountering a specific bacteria/virus
 Initiated by **ANTIGENS**
 $\downarrow$
 Antibody generators

2 types:

Humoral - B-cell
 B lymphocyte Attack

Cell mediated - T-cell
 T lymphocyte Destroy

RBC

People $\in$ vitk deficiency are probably excessive bleeders.

Clotting factors produced in Liver by hepatocytes
EXCEPT Factor 8 (produced in Liver but Not by hepatocytes)

intrinsic = SLOW
 $\odot$ changed sub-endothelial substance/vascular injury

Extrinsic = Explosive
 TF

usually clotting factors = inactive but $\sim$ 1% Factor 7 is the exception (spontaneously activated)

AT3 inhibits Thrombin & Factors 7, 9, 11, 12
 extrinsic $\rightarrow$ intrinsic

Heparan sulfate makes AT3 much STRONGER!

Ventilation: Breathing

Gas exchange

- External Respiration - Ventilation/gas exchange in the lungs
- Internal Respiration - oxygen utilization & gas exchange b/w blood & tissues

occurs via **Simple diffusion**

- O_2 concentration higher in lungs than blood
- CO_2 concentration higher in blood than lungs

Respiratory system:

Conduction zone - transports air → trachea + bronchi → warms, humidifies, filters & cleans air; mucus traps particles

Respiratory zone - site of gas exchange terminal bronchioles; **Alveoli**

Alveoli: Large SA for ↑ diffusion rate

Type 1 - SA for gas exchange

Type 2 - Secrete **Surfactant** prevents alveoli collapse (reduces surface tension)

Elasticity: Lungs return to initial size & stretch (Recoil)

Lungs attached to thoracic wall = Always under **elastic tension**

→ Inc during inspiration
dec during expiration (by elastic recoil)

Breathing:

Inspiration - Diaphragm contracts/flattens
Accessory: Serratus, pec minor, SCM
External intercostals

Expiration - Quiet - Diaphragm relaxes
Forced - Internal intercostals, Abdominal muscles

	Inspiration (↓ intrapulm pressure)	Expiration (↑ intrapulm pressure)
Quiet breathing	-3	+3
Forced ventilation	-20	+30

*So as volume goes ↑, pressure goes ↓, v/v ↑

Breathing Regulation:

Voluntary control:

Cerebral cortex

Involuntary control:

Medulla & pons

Apneustic center = influences medulla

Inspiration

Pneumotaxic center =

Expiration

Chemoreceptors:

monitor pH & CO_2 levels

Feedback controls

automatic breathing

Increased CO_2 is the primary drive for breathing

ventilation controlled to maintain constant CO_2 levels in the blood (pH 7.35-7.45)

Gas Exchange:

Dalton's Law:

total gas pressure = sum of pressures of each gas in it

Partial pressure:

pressure of individual gas; % of gas x total pressure

PO_2 ; PCO_2

Determines Gas Exchange Direction

Only measures O_2 dissolved in blood plasma - Does Not measure O_2 bound to hemoglobin in RBCs

When functioning properly:

PO_2 of systemic arterial blood is only 5 mmHg less than

PO_2 of alveolar air → PO_2 Arteries = 100 mmHg

PO_2 Alveoli = 105 mmHg

hyperventilation = ↑ CO_2 → ↓ pH

HYPERVENTILATION = ↓ CO_2 → ↑ pH

hypocapnia

O_2 level indirectly affects ventilation by affecting chemo-receptor sensitivity to CO_2

Ex. LOW O_2 = Carotid bodies become more sensitive to CO_2

Hypoxic drive: Carotid bodies respond directly to low O_2 dissolved in plasma

Pressure types:

Atmospheric - pressure outside body

Intrapulmonary - pressure inside lungs

Intrapleural - pressure within intrapleural space
(-) pressure to keep lungs from collapsing
b/w parietal & visceral layers
contains thin fluid = lubricant

Boyles Law: pressure inversely proportional to volume

When breathing:

INSPIRATION -

Intrapulmonary pressure LOWER than

Atmospheric pressure

EXPIRATION -

Intrapulmonary pressure greater than

Atmospheric pressure

→ Increase in lung vol then decreases intrapulm pressure
≡ Air goes in

→ Decrease in lung vol then increases intrapulm pressure
≡ Air goes out

Law of Laplace: pressure inversely proportional to alveoli radius

Large radius = lower pressure

Surfactant important here:

reduces surface tension; prevents collapse

Allows residual vol of air to stay in lungs

Lung Vol & Capacities:

Tidal Vol (TV) - Normal breathing volume (quiet breathing)

Inspiratory Reserve Vol (IRV) - Forceful inhaled above TV

Inspiratory Capacity - Amt that can be inspired & TV
TV + IRV

Expiratory Reserve Vol (ERV) - Forceful exhaled below TV

Residual Vol (RV) - Air left in lungs & max expiration

Functional Residual Capacity (FRC) - Air left in lungs & Normal expiration
ERV + RV

Vital Capacity (VC) - Max amt that can be forcefully exhaled & max inhalation
VC = RV

Total Lung Capacity (TLC) - Amt of all gas vol in lungs
IRV + TV + FRC
IC + ERV + RV

Stimulus

↑ PCO_2

↓ pH

↓ PO_2

Chemoreceptor

Medullary & Aortic & carotid bodies

Aortic & carotid bodies

Carotid bodies

Respiration Cont.

Hemoglobin = carries O_2 in the blood
 Each RBC contains millions of hemoglobin
Structure: Each hemoglobin molecule has 4 subunits; each subunit binds to one molecule of O_2

Oxyhemoglobin Saturation $\rightarrow$ measurement of how well lungs have oxygenated blood
 % oxyhemoglobin to total hemoglobin

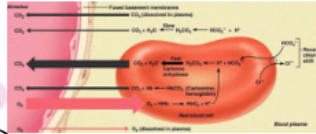
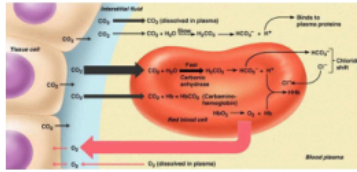
Hemoglobin concentration $\rightarrow$ measures oxygen-carrying capacity of blood

Anemia $\rightarrow$ low hemoglobin concentration;
 = reduced O_2 transport capacity

Polycythemia $\rightarrow$ high hemoglobin concentration;
 can be caused by high altitudes

Erythropoietin = produced by kidneys; stimulates hemoglobin/RBC production in response to low O_2 levels

@ high altitudes
 O_2 supply goes $\downarrow$
 causing more hemoglobin production
 = polycythemia



Carbon Dioxide transport:
 carried in blood by 3 forms $\rightarrow$
 1) Dissolved in plasma (CO_2) 7-10%
 2) Carbaminohemoglobin 20%
 (CO_2 attached to Hb in hemoglobin)
 3) Bicarbonate ions (HCO_3^-) Majority

Chloride Shift $\rightarrow$
 Exchange of HCO_3^- and Cl^- across the RBC membrane to maintain ionic balance
 * Bicarb leaves RBC
 * Chloride diffuses in
 Charge remains the same.

@ tissues: $CO_2 + H_2O \xrightarrow{CAD} H_2CO_3$
 $H_2CO_3 \rightarrow H^+ + HCO_3^-$
 Reverse Chloride Shift
 Chloride Shift

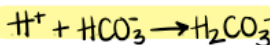
@ lungs: $HCO_3^- + H^+ \rightarrow H_2CO_3$
 $H_2CO_3 \xrightarrow{CAD} H_2O + CO_2$

Reverse Chloride Shift $\rightarrow$
 In pulmonary capillaries
 $\uparrow P_{O_2}$ favors production of oxyhemoglobin

So H^+ dissociates from hemoglobin
 $\&$ recombines $\bar{c}$ bicarb (HCO_3^-)
 to form carbonic acid (H_2CO_3)

Chloride (Cl^-) diffuses out of RBC and bicarb (HCO_3^-) enters

inside the RBC is where H^+ binds $\&$ CAD converts



* Influenced by pH, temp, $\&$ partial pressure of O_2

O_2 loading: O_2 binds to hemoglobin in the lungs

O_2 unloading: Oxyhemoglobin drops off O_2 in the tissues

Dissociation Curve:
 97-75 = 22
 22% O_2 is unloaded to tissues

Veins: P_{O_2} of 40 mmHg
 $\rightarrow$ enough O_2 bind for 75% oxyhemoglobin
Arteries: P_{O_2} of 100 mmHg
 $\rightarrow$ enough O_2 bind for 97% oxyhemoglobin

Curve Shifts = influenced by pH + temp

Bohr Effect: describes effect of pH on affinity of hemoglobin for O_2

Higher pH's = hemoglobin binds more readily
 Lower pH's = Offloading O_2 ; wants to get rid of it

Alkaline / More Basic:
 Increased affinity for O_2 $\rightarrow$ better binding in lungs (leftward shift)
 pH $\uparrow$, Affinity $\uparrow$

More Acidic:
 Decreased affinity for O_2 $\rightarrow$ increased offloading to tissue (rightward shift)
 pH $\downarrow$, Affinity $\downarrow$

Lower temp:
 Increased affinity for O_2 $\rightarrow$ better binding in lungs (leftward shift)
 Temp $\downarrow$, Affinity $\uparrow$

Higher temp:
 Decreased affinity for O_2 $\rightarrow$ increased offloading to tissue (rightward shift)
 Temp $\uparrow$, Affinity $\downarrow$

Note:

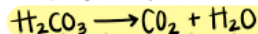
Bonding of H^+ to hemoglobin lowers affinity for O_2

$\rightarrow H^+$ helps blood carry more CO_2

Ex. $\downarrow$ pH = $\uparrow$ H^+ concentration = $\downarrow$ O_2 affinity resulting in inc O_2 offloading

@ the alveoli:

Carbonic Anhydrase (CAH)
 Converts Carbonic Acid back into carbon dioxide + water



CO_2 is exhaled (through lungs)

Acid-Base regulation:

Maintained by →

Lungs + **Kidneys**

quick Δs

Slow Δs

Regulates volatile acids (by breathing)

Regulates NON-Volatile Acids

Ex: Carbonic Acid (H_2CO_3) → because it can be converted into gas & exhaled

Ex: lactic Acid, Ketones, etc → Buffered by bicarb (HCO_3^-) but cannot be regulated by breathing

* Constant range to be maintained: 7.35–7.45

↓
If kidneys detect low HCO_3^- in blood, they will reabsorb vs.
If they detect high HCO_3^- in blood, they will excrete it

Acid-Base Disorders:

Acidosis → pH Below 7.35

Respiratory Acidosis

caused by hypoventilation
Rise of CO_2 (hypercapnia)
 $\uparrow CO_2 \rightarrow \uparrow H^+ \rightarrow \uparrow H_2CO_3$
Result = pH decreases

$P_{CO_2} > 40$

Metabolic Acidosis

caused by excessive Acid production ↑ NonVolatile Acids
Loss of bicarb (HCO_3^-) = Diarrhea

$HCO_3^- < 24$

Alkalosis → pH ABOVE 7.45

Respiratory Alkalosis

caused by Hyperventilation
Fall of CO_2 (hypocapnia)
"Blow Off" = fast breathing
 $\downarrow CO_2 \rightarrow \downarrow H^+ \rightarrow \downarrow H_2CO_3$
Result = pH increases

$P_{CO_2} < 40$

Metabolic Alkalosis

caused by inadequate Acid production ↓ NonVolatile acids
over production of bicarb (HCO_3^-) = loss of digestive acids via Vomiting

$HCO_3^- > 24$

Compensation:

Lungs
Adjust ventilation to alter CO_2 levels

Kidneys adjust bicarb reabsorption or excretion

Normal blood pH =

maintained when ratio of bicarb to CO_2 is 20:1

Henderson-Hasselbalch Equation:

$$pH = 6.1 + \log \frac{HCO_3^-}{(0.03)P_{CO_2}}$$

Effects of Exercise + High Altitude →

Exercise:

↑ O_2 demand + ↑ CO_2 production results in deeper + faster breathing (tachypnea)

Arterial P_{O_2} + P_{CO_2} remains stable despite the increased activity

Altitude: ↓ P_{O_2} at higher altitudes affects oxygen loading →

3 Acclimation Adaptations:

1) Immediate – ↑ ventilation

hyperventilation lowers P_{CO_2} (respiratory Alkalosis)

↑ affinity of hemoglobin for O_2 = Better O_2 loading

Leftward Curve Shift

2) Days – ↑ 2,3 DPG

↑ affinity of hemoglobin for O_2 = Increase unloading to tissues

Rightward Curve Shift

3) weeks – ↑ EPO = ↑ RBCs

= Increased O_2 content in blood

Ventilation can compensate for the metabolic component:

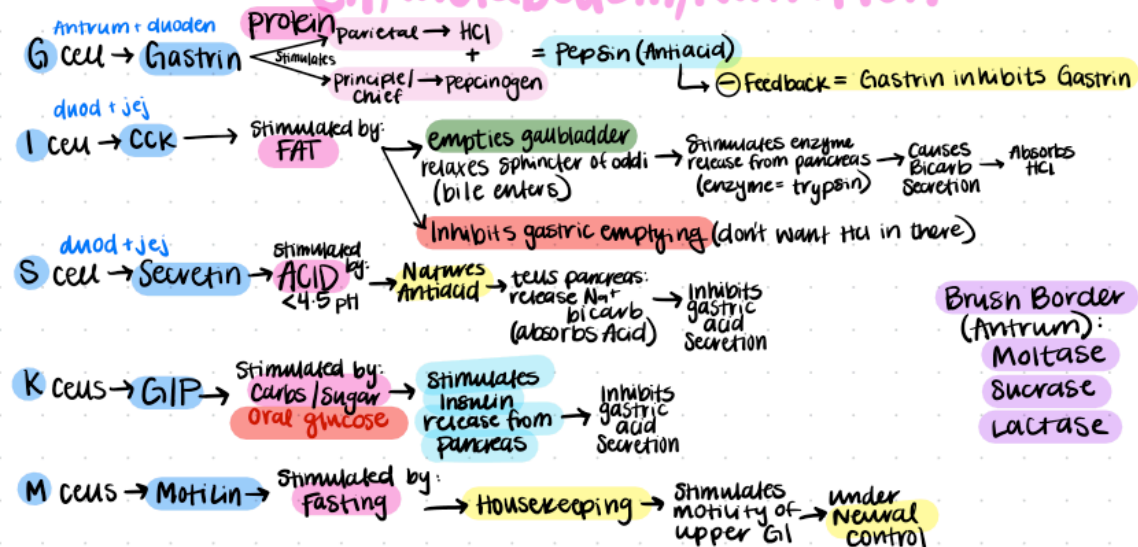
Condition	pH	P_{CO_2}	Ventilation	Cause or Compensation
Normal	7.35–7.45	35–45 mmHg	Normal	Not applicable
Respiratory acidosis	Low	High	Hypoventilation	Cause of the acidosis
Respiratory alkalosis	High	Low	Hyperventilation	Cause of the alkalosis
Metabolic acidosis	Low	Low	Hyperventilation	Compensation for acidosis
Metabolic alkalosis	High	High	Hypoventilation	Compensation for alkalosis

Have to determine if ventilation rates are the cause or the compensation

* Pay attention to P_{CO_2}

Low pH = Always Acidosis

High pH = Always Alkalosis



Brush Border (Antrum):

- Maltase
- Sucrase
- Lactase

paracrines

peptide - **Somatostatin** (stops all gut hormones)

non peptide - **Histamine** (stimulates Acid secretion)

Endocrine

hormones; Act on distant target cells

Neurocrine

peptide = **VIP**

non peptide = **NE | Epi, Ach**

PANCREAS

Endocrine + Exocrine gland

cells located in the pancreatic islets.

α cells = glucagon
β cells = insulin

helps mobilize glucose in bloodstream to help ↑ blood sugar

helps to lower blood sugar into normal range

Primary hormone regulating plasma glucose concentration

Detected in the Liver

Insulin helps glucose get into tissue cells

Insulin is anabolic hormone
→ causes tissue growth

- Once insulin binds @ cell surface — glucose transporters translocate to cell surface — which allows glucose to come in

* Important because normally cells are impermeable to glucose

Important tissues:

Muscle
Liver
Adipose tissue

Diabetes -

Type 1: Lack of insulin

Type 2: insulin resistance (due to ↑ blood sugar for so long)

- ① binds to receptors on target cells
- ② indirectly stimulates enzyme glycogen synthase to promote sugar storage (muscle + liver)
- ③ stimulates fat storage (adipose tissue) for long-term storage

Glucagon: Antagonist to insulin

Secreted when blood glucose levels are low

Detected in the Liver, Secreted from Pancreas

- ① Stimulates liver to hydrolyze glycogen into glucose
- ② releases into blood
- ③ Gluconeogenesis stimulated: conversion of noncarbs → glucose
- ④ Stimulates lipolysis in adipose tissue → fat released & used as fuel source (instead of glucose)

ANABOLIC

↑ Blood Sugar → α insulin → ↓ Blood Glucose to storage

CATABOLIC

↓ Blood Sugar → β glucagon → ↑ Blood Sugar from liver storage

KIDNEY

Secretes Erythropoietin (EPO)

Stimulates production of RBC by red bone marrow (flat bones - Sternum, Rip, & Vertebrae)

PINEAL

Secretes Melatonin

Regulated by the hypothalamus

- Stimulates production when it gets dark
→ regulation of circadian rhythms
- production inhibited during day time

THYMUS

Secretes thymopoietin

increases WBC production in lymph nodes

Hormones help regulate body metabolism growth + reproduction

Antagonistic - hormones working in opposite direction
Ex: insulin / glucagon

Synergistic - work together to produce effect
Ex: prolactin / oxytocin

Permissive - one hormone making target cell more responsive to 2nd hormone
Ex: Estrogen + progesterone

Hormones have 1/2 life time to lose 50% of something. Usually hormones are removed from the blood by the liver

Higher plasma concentrations of hormones causes effects different from normal

Body response - prolonged exposure → decreased # of hormone receptors = Downregulation

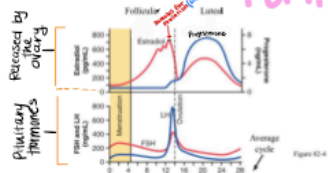
Fewer receptors = fewer hormones can bind = less effect/response

Protein mechanism

up regulation / priming effect
if we give drug that ↓ activity of receptors the body will increase the # of receptors to become more sensitive & be more likely to bind to target hormones

REPRODUCTION

FEMALE REPRODUCTION



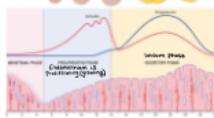
key:

- Corpus Luteum = Yellow
- Corpus Albicans = Red
- FSH = Causes follicle to mature
- LH = Causes egg to be expelled (ovulation)

- ① **FSH** secretion causes follicle to mature
- ② As follicle matures it secretes **Estrogen**
- ③ **Estrogen** increasing triggers quick increase of **LH + FSH** (but mostly **LH**)
- ④ **LH** peak surge induces ovulation =
Ovum expelled from follicle
- ⑤ **LH** surge also stimulates production of the corpus luteum
- ⑥ Corpus luteum is the only thing that produces **progesterone**
↳ Also produces some levels of **estrog**
- ⑦ **Progesterone** prepares uterus endometrium for implantation
- ⑧ **Estrogen** imposes **○** feedback on **FSH**
↳ released into bloodstream & tells **FSH** to slow down



Males:

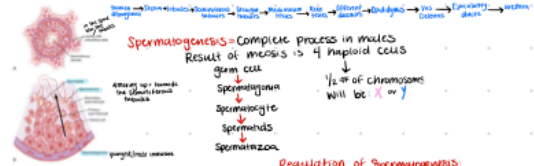
[illegible]

Progesterone causes big increase in Endometrium

* side note: corpus luteum degenerates
3. becomes corpus albicans
 ↓
 scar tissue

- ① **Primordial** - Single layer of granulosa cells
- ② **Primary** - Multiple layers of granulosa cells
Ovarian enlargement, Regulated by LH, FSH
- ③ **Antral** - Fast granulosa cell proliferation
→ producing follicular fluid
→ spindle cells develop into theca
- 
- Theca interna: Secrete sex hormones
estrogen, progesterone
- ④ **Corpus luteum** - degenerates into corpus albicans b/c there is no FSH to maintain it
- ⑤ **No more corpus luteum** = No more **progesterone** + **estrogen**
the decline allows FSH secretion to start back up
- ⑥ **Menstruation**
- ⑦ **Repeat**

male reproduction



Regulation of Spermatogenesis:

- LH:** Stimulates Leydig cells to secrete testosterone
- Testosterone:** ↑ growth/division of germinal cells
- FSH:** Stimulates Sertoli cells (support for sperm)
- Estrogens:** made from testosterone; FSH stimulates Sertoli
- GH:** Promotes division of spermatogonia
↳ Needed for sperm production
- Inhibin:** Produced by Sertoli cells
⊖ feedback regulation of FSH

Composition of Ejaculate

- Semen = Basic pH → Sperm don't like Acidophylic acidic environments

Capable of motility within 18-24 hr of
Storage in the epididymis

Seminal vesicles - mucus containing
fructose, citric acid,
prostaglandins, fibrinogen

Prostatic gland - Milky fluid containing Ca^{++} , creatine ion, phosphatidase and proflimindolysin

Semen - Fluid + Sperm, pH - 7.6

• (vaginal pH is 4)

Sperm
Mature after ejaculation:
400 million sperm / ejaculate
Storage up to 1 month in
the epididymis
Likes: Neutral to basic pH &
2°C under body temp

Testosterone important for:
 Enlargement of primary sex characteristics
 Hair growth deep voice ↑ skin thickness
 ↑ bone density / Too much? Baldness
 production starts @ about 10 years

Estrogen Effects:

- ↑ Cell proliferation
tells vaginal epithelium
to become stratified
(from cuboidal state)
- ↑ Uterus size
- ↑ Ciliated epithelium
of fallopian tubes
- ↑ Stroma breast tissue
(lead to tenderness)
- ↓ Osteoclastic activity
- Tells cervix to secrete
watery soln (so sperm can swim)

Progestational effects

- ↑ Secretory Δ in endometrium
- ↓ Uterine Contraction
Egg should already be fertilized
- ↑ Secretion in fallopian tube
- ↑ Breast Ischemia + Atrophy
- Tells cervix to secrete a thick viscous secretion
- ↑ body temp by 0.5°C

Puberty
gradual ↑ GnRH in 8th year
onset b/w 11-16 years old

Menopause 1st cycle
Avg onset ~ <13 years old

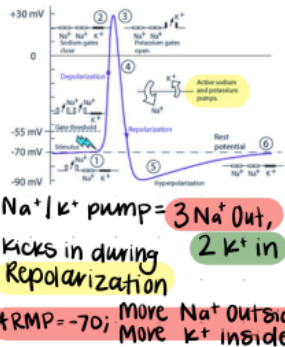
(- gonadotropin)
↓
↓ inhib PSH
due to no
⊖ feedback

No
RELINQUISH

Sensory
Afferent signals → Spinal cord → Brainstem → Cerebellum → Thalamus → Cerebral Cortex

Motor
Efferent signals → CNS → Muscles/Glands

Levels of control:
Lower regions = automatic
Higher regions = deliberate
Major levels of CNS function:
Spinal cord level = signal transmission / reflex arcs
Lower brain level = (subcortical / lizard brain) subconscious activities i.e. breathing, emotional patterns
Higher brain level = (cortical) memory storehouse, conscious control, critical thinking



Na^+/K^+ pump = 3Na^+ Out, 2K^+ in
Kicks in during Repolarization
RMP = -70; More Na^+ Outside, More K^+ Inside
ARP = During AP, Na^+ channels inactive (Not just closed)
RRP = K^+ channels still open, Strong stimulus can overcome this (i.e. seizures)

Synaptic excitation:
Summation
Spatial → Multiple neurons sum up to 1 AP
Temporal → many signals from ONE neuron until threshold met
Inhibitory neurons @ the axon have greater effects than excitatory neurons further away

Neuroglia
CNS 1 to many
Oligodendrocytes form myelin sheath = quick AP transmission
Microglia phagocytize foreign & degenerated material = CNS immune function
Astrocytes Regulate external environment / bouncer = Regulates the BBB
Ependymal line the ventricles & secrete CSF
PNS
Schwann cells many to 1 form myelin sheath Phagocytic activity = infection prevention
Satellite cells support cell bodies, regulate external chemical environment = structure + support
Selective permeability to protect brain
Disease association: Meningitis
Myelin sheath gives axons their white color = white matter
Grey matter = cell bodies + dendrites = NO myelin sheath
Disease Association: Multiple Sclerosis = Axons Striped of myelin sheath

Most Memory Storage happens here
↓
Repetitive synapse firing solidifies memories
APs = Nerve Impulses
Types of Synapses →
Electrical: Bidirectional transmission via gap junctions (NOT as common)
Chemical: Unidirectional transmission
Neurotransmitters that excite or inhibit
↓
Synaptic vessels contain the neurotransmitters
Voltage-gated Ca^{2+} channels
More Ca^{2+} = More vesicle release
Synapse = Allows mvm of ions
Receptor proteins: Gated ion channels
Cation - Excitatory Channels $\text{Na}^+, \text{K}^+, \text{Ca}^{2+}$
Anion - Inhibitory Channels Mainly Cl^-

	Synthesis	Metabolism	Uses	Receptors
ACh	Acetyl CoA + Choline (choline acetyltransferase)	Acetylcholinesterase (enzyme recycled by presynaptic neuron)	Somatic & Autonomic (SNS & PNS)	Cholinergic: Nicotinic & Muscarinic
NE	Made from tyrosine (in diet: cheese, meat)	COMT, MAO	SNS (usually excitatory)	Adrenergic (α, β, P, B_2)
Dopamine Reward Can't pass BBB	Made from tyrosine (in diet: cheese, meat)	COMT, MAO	Substantia Nigra secretes Nigrostriatal - Motor Mesocortical - Reward Tuberoinfundibular - Outputs to pituitary	D_1 - stimulates D_2 - inhibits
Serotonin 5-HT Feel good	Made from tryptophan	MAO	Alters mood/role in Anxiety, Nausea, Vomiting, Migraines	Many
GABA Brain = car GABA = brakes	Made from Glutamate (+ B6)	GABA transaminase	Major INHIBITORY neurotransmitter hyperpolarizes postsynaptic neurons	GABA-A (opens Cl^- channels) = cell more $\ominus$ = harder to have AP
Glutamate	Made from Glutamine	GABA	Major EXCITATORY neurotransmitter Excitotoxicity can occur & overstim Alzheimers; Cell suicide	NMDA, AMPA Metabotropic receptors

protective mechanism:
Fatigue - prevents neuronal damage
Important for seizure termination

Effects of pH: ex. hyperventilation (panic attacks) → F/F response → ↓ CO_2 → Blood pH ↑
Alkalosis ↑ excitability
Acidosis ↓ excitability
pH > 7 = Alkalotic

Others:
Glycine - inhibitory in Spinal cord
Histamine - H1 receptor Effects arousal (Ex. Benedryl = histamine inhibitor, makes you sleepy)
Nitric oxide - behavior & long term memory
Cannabinoids - CB1, CB2 receptors, mood, appetite, nausea, vomiting, cognition, memory

Neuro Intro Cont.

Oxygen = high requirement for neuronal excitability
 Hypoxia / No cerebral bloodflow for 3-7s leads to unconsciousness

Receptor potentials: Frequency of repetitive APs determine sensation

→ one signal may not be enough, but if many signals coming in → brain gives it priority

Phasic receptors =

good for sensing Δ ; fast adapting
 ex. feeling clothes you wear

Tonic receptors =

Slow adapting; maintain high firing rate as long as stimulus is applied

ex. hand on stove - continuous signal until stimulus removed

baro/chemo receptors in CV system: respond to Δ in pTT / Δ in CO_2

Degree of Accomodation is related to the type of tissue stimulated

Fast pain

< 0.1s $\bar{P}$ stimulus

Sharp, Acute
 Skin only

↓
 Neospinothalamic tract
 tactile receptors in dorsal column tract = Hand on stove
 • Release glutamate (Excitatory!)

Analgesia System:

Natural Anti-pain pathway
 • Serotonin → Stimulates Enkephalin release
 Reduce pain signals
 Natural Opioid

Slow pain

> 1s $\bar{P}$ stimulus

Aching, chronic
 Skin + deep tissue

★ Substance P involved

Nociceptors adapt very little & can become more sensitive overtime = **Hyperalgesia** (more easily triggered)

↓
 Paleospinothalamic tract:

Poor ability to locate
 • Release glutamate + Substance P helps sensitize nerves to feel pain more acutely

Sensory categories:

Chemoreceptors - **Chemicals** (also in taste & smell)
 Photoreceptors - **Light**
 Thermoreceptors - **Temperature**
 Mechanoreceptors - **Mechanical deformation** (hearing, touch)
 Nociceptors - **Pain** (temp, pressure, chemicals)
 Proprioceptors - **Mmt/balance** (muscle, tendons, joints)
 Special senses - **Equilibrium** (vision, hearing, taste, smell)
 Cutaneous receptors - **Skin**

Merkel's discs: Indentation

Root hair plexus: Light brushing

Pacinian Corpuscle: Vibration; deep pressure

Meissner's corpuscle: Change in texture

Free nerve ending: touch, pain, hot, cold

Ruffini ending: Skin stretching

Sensory Pathways:

Dorsal Column-Medial Lemniscal System:

Fast transmission

High Spatial Orientation
 touch detection & high locality & fine gradation of intensity

Crossover in Medulla

Think: fingertips

Anterolateral System:

Slower transmission

Detects broad sensory

Spectrum-

Pain, temp, crude touch

Crossover in Spinal column

Think: tickle/itch

Somatosensory areas →

Areas & larger areas in somatic cortex have larger # of sensory receptors

Greater density of nerves in sensitive areas

= face + hands

Lower 2pt touch threshold

Less sensitivity in larger areas

= back + abdomen

higher 2pt touch threshold

Recruitment process for signal intensity

Lateral inhibition for fine differentiation

Thermal Sensations:

Cold receptors more numerous

Nociceptors detect extreme hot or cold

Temp Δ respond more than static temp

Visual pathway:

Cornea → Anterior chamber → pupil → posterior chamber → lens → posterior cavity → Retina

Iris = outer radial muscle + inner circular muscle
Can increase/decrease diameter of the pupil (CN III)

Constriction
Parasympathetic stimulation
Circular muscle contract

Bright light response

Dilation
Sympathetic stimulation
Radial muscle contract
MYDRIASIS

Dim light response

fovea = most cones
optic disc = blind spot
peripheral vision = more rods

Retina = where the photoreceptors live

Located in the pigmented choroid (dark/contains melanin) - good b/c this means it helps to absorb lots of light

Lens = Avascular
Sits behind the iris

lens is attached to suspensory ligaments (zonules)

which are controlled by: ciliary muscles/bodies

These produce: **Aqueous humor**

which fills the Ant/post chambers of the Ant cavity provides nourishment to lens + cornea

Drains into the Canal of Schlemm & then back into blood

The lens helps to bend the light coming in:

Light Refraction - image is flipped upside down & Left to Right

Lens Accommodation:
Keeps objects focused on the retina as the distance b/w eye & object moves

FAR VISION

Ciliary muscle relax
Suspensory ligaments tighten
lens thins & flattens

Sympathetic

NEAR VISION

Ciliary muscle contract
Suspensory ligaments relax
lens thickens & rounds

parasympathetic

Myopia = Nearsightedness

Elongated eyeball

Distant images brought to focus in front of the retina

Correction: **Concave lens**

Hyperopia = Far sightedness

Short eyeball

Distant images brought to focus behind the retina

Correction: **Convex lens**

Snellen eye chart measures visual acuity: near point / sharpness of vision / resolving power

Astigmatism = irregular curvature

Several points of focus on the retina

Correction: **cylindrical lens**

Presbyopia = age-related

reduced lens flexibility

Cataracts = cloudy lens

Correction: **surgical removal**

Visual pathway:

Retina → pigmented cells
photoreceptors

Horizontal cells - send signals laterally for an inhibitory effect: when focusing on one spot, will inhibit other points

bipolar cells

Amacrine cells

Ganglionic cells

Optic nerve
CN II

Rods: B & W vision in low light

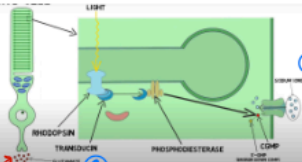
contain **Rhodopsin** (purple pigment)

which absorbs green light best

* Requires **Vitamin A** *

Phototransduction:

occurs in Retina - light is converted into electrical signals



Na⁺ inflow causes continuous depolarized state = continuous release of glutamate (excitatory ntm)

Rhodopsin is made up of retinal - which is what absorbs light (Vit A necessary)

when light is absorbed, protein transducin is activated

which then activates phosphodiesterase

Cones: Blue, Green, Red

Color blindness - X-linked

Loss of **Red** cones: **Protanopia**

Loss of **blue** cones: **tritanopia**

Loss of **green** cones: **Deutanopia**

ion channels begin to close

phosphodiesterase breaks down cGMP

Cell becomes hyperpolarized (K⁺ leaving, No Na⁺ entering)

glutamate release decreases

Acts as a signal that a light stimulus is present

19,20 Vision / Senses cont.

more on color blindness:
failure of 2 types of cones
= **Monochromacy**
To detect color blindness -
Ishihara test

Eye movement:
LRG SO4(3)

Fixation movements:

Voluntary = fixate on chosen point
frontal cortex control

Involuntary = maintain fixation
occipital cortex control

Saccadic Movement = quick, random movements to keep new visual inputs

Converts sound vibs into electrical inputs

Stapedius muscle dampens Stapes if sound is too loud

Inner ear = the cochlea

Scala Vestibuli
Portion of bony labyrinth

Scala Media (Endolymph)
Cochlear duct

Scala tympani

Hearing Impairment:

Conductive: Wax, fluid, Otosclerosis (bone overgrowth)

Sensorineural: hair cell damage

Presbycusis: age-related

Smell:

CNI olfactory nerve = Short transmission to brain

Olfactory receptors - bipolar neurons & ciliated dendrites projecting into nasal cavity.

Proteins bind to odors

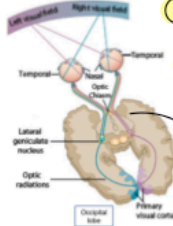
1 odorant binds to 1 protein (specific receptors)

Olfactory nerves = unmyelinated

Olfaction greatly influences taste

Interactions & the **thalamus** & **limbic system**

Visual pathway:



① visual field is projected onto the **R** side of the retina in both eyes - then gets transferred to the **R** lobe of the occipital cortex

Medial aspects crossover at the **optic chiasm**

Optic nerve → optic chiasm

Lateral geniculate nucleus (THALAMUS) ← Optic tract

Primary Visual Cortex
(occipital lobe of cerebrum)
processes initial visual information

Secondary Visual Cortex
integrates motion 3D position, color, and details

Hearing:

Middle ear = b/w TM and cochlea

contains 3 bones (ossicles)

• **Malleus** → vibrations transmitted along the bones
• **Incus**
• **Stapes** → attached to oval window

High frequency = terminate early
Low frequency = travel further

Sound waves enter **Scala media**

tectorial membrane vibrates

Stereocilia bends

K⁺ returns to perilymph @ base of the stereocilia

K⁺ rushes in = **Depolarization**

K⁺ channels facing the endolymph **OPEN**

contains organ of Corti = converts sound waves into electrical impulses = **vestibulocochlear nerve (CN VIII)**
Contains **hair cells** - located on **basilar membrane**, projecting into endolymph of cochlear duct

Taste: gustation
receptors = taste buds
taste pathway:
CN VII + CN IX

Medulla oblongata → Thalamus

gustatory cortex

taste Receptors:

Salty = Na⁺ depolarizes
Sour = H⁺ depolarizes

Needs Ca²⁺ to release ntm

Sweet & Meat = sugar/glutamate binds & activates G-proteins → 2nd messenger close K⁺ channels

Bitter = quinine binds & activates G-protein → 2nd messenger releases Ca²⁺ into cell

Swallowing coordinated by:
CN V VII IX X XI XII
5 7 9 10 11 12

types of papillae:

Fungiform CN VII Facial

Anterior surface

Circumvallate CN IX glossopharyngeal

Posterior surface

Foliate/Filiform CN IX glossopharyngeal

Sides

Salivary glands:

Sublingual CN VII Facial

Submandibular CN VII Facial

Parotid CN IX glossopharyngeal

G-protein coupled ATP odorant molecule binds to receptor protein → Activates **cAMP**
cAMP opens Na⁺ & Ca²⁺ channels = graded **depolarization** → AP
up to 50 g-proteins associated & 1 receptor protein - gives great sensitivity through **amplification**

Interneurons = found in cord grey matter

Integrate spinal cord function

Sensory signals/signals from brain
more likely to terminate on interneurons

Responsible for sending motor message out

Renshaw cells → focusing of signals
Propriospinal fibers → coordinate mvmt in various areas/limbs

Muscle Spindles Stretch detectors **Golgi tendon organs** = tension detectors

Central region has NO Actin or Myosin

Unable to contract by itself

Sensory primary - encircles center of fiber
Secondary - 1-2 smaller fibers branching out

Stretch → firing
Shortening → firing

These occur when muscle length is kept near its resting length
Dynamic: Sudden Δs
Primary: Fibers stimulated

Static: Slow & Steady Δs
Primary & Secondary: Fibers Stim.

Reflex Arc:

Quick heavy book dropped into arms
Muscle Spindle STRETCHED → Sensory firing ↑

Causes contraction of same (or same) muscle(s)

Spindle no longer stretched

Afferent to spinal cord

Inhibitory interneuron branches off

Sends inhibitory motor signal to (inverts) antagonist muscle

Causing relaxation

Flexor reflex = Nociceptor reflex (elicited by pain) most powerfully

*quick! happens within spinal cord

Voluntary Activity:

α-γ co-activation =

Initia + extrafusal fibers contract @ same time

Prevents receptor (sensory) from activating & causing reflex arc

Maintaining dampening fcfn

motor neuron causes muscle to contract (α)

Muscle spindle goes slack - No sensory sensitivity

AS extrafusal fibers are pulled apart, intrafusal fibers are stimulated & they oppose the reflex

Central control of γ motor neurons helps to stabilize joints → good for anti-gravity muscles & muscle movements under load

Cerebellum → "Silent area"

Useful for smooth progression

Makes corrective adjustments

Plans sequential mvmts

learns mistakes = muscle memory

★ Vermis - Axial body

★ Intermediate - distal upper & lower limbs

★ Lateral zone - joins & cerebellar cortex

Assists in planning sequential mvmts (i.e. dancing)

Neuronal circuits of cerebellum:

Where are signals coming from? **Input pathways**

Corticopontocerebellar pathway - from areas of brain making the decisions

Olivocerebellar tract - from motor cortex, basal gang, spinal cord

Vestibulocerebellar fibers (detecting mvmt/equilibrium) - vestibular apparatus

Reticulocerebellar fibers - brainstem reticular formation

Sensory Afferent pathways from periphery -

DORSAL Spino-cerebellar tract - tells cerebellum what's going on in body

muscle contraction, tendon tension, positions, forces

VENTRAL Spino-cerebellar tract - tells cerebellum what signals have been

received by Ant. horns (which go on to activate mvmt)

Primary Motor Cortex:

Stim of neurons excites patterns of separate muscles

Premotor Area:

Stim activates complex patterns of mvmt

Develops "motor image" = sends to PMC

or basal ganglia to be carried out

Supplementary Motor area (side 20)

Stim causes bilateral mvmts

WIPs & premotor area for "positioning"

head rotation, eye mvmt, etc

Sensory fibers feed into Motor Cortex to tell it what they want to happen.

Red Nucleus:

Alternative pathway - Cortical signals → spinal cord

Skips cerebellum = basal gang

Used for discrete activation

of tiny muscles (fingers/hands)

(Small in humans bc we don't use it)

Transmission of signals to muscle corticospinal or pyramidal tract terminate on interneurons
Accessory pathways: Basal ganglia Cerebellum

Brainstem:

Medulla, pons, mesencephalon

Control involuntary functions

helps control signals that make it back down to the muscles

Pontine reticular nuclei = Excitation of antigravity muscles

Medullary reticular nuclei = Relaxation (ANTAGONISTIC)

Vestibular Nuclei: Detects Stimulus & reports to Pontine reticular nuclei to maintain equilibrium

Someone Pings your toilet → Vestibular Stimulus → Pontine reticular Nuclei Activated → Antigravity muscles stimulated → helps you stand

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Cerebellum strengthens signals from cortex ex. coordinates on/off for antagonistic muscles